



CS 5384-001/D01

Logic for Computer Scientists

Lecture 21

10/29/25



Nested Quantifiers

- $\forall x \forall y P(x, y) = T$, if $P(x, y) = T$ for all x and for all y .
- $\forall x \exists y P(x, y) = T$, if for all x there exists y such that $P(x, y) = T$.
- $\exists x \forall y P(x, y) = T$, if there exists x such that $P(x, y) = T$ for all y .
- $\exists x \exists y P(x, y) = T$, if there exists x and there exists y such that $P(x, y) = T$.

How Do the Quantifiers Translate?



Assume $P(x, y)$ means x likes y , where

$x \in \{\text{John, Jack, Jeff, etc.}\}$ - A student

$y \in \{\text{Pizza, Sandwich, Salad etc.}\}$ - A fast-food item

- $P(\text{John, Pizza})$: John likes pizza
- $\exists y P(\text{John, } y)$: John likes a fast-food item
- $\exists x (P(x, \text{Pizza}) \wedge P(x, \text{Salad}))$: There is a student who likes both pizza and salad
- $\exists x \exists y \exists z ((x \neq y) \wedge (P(x, z) \rightarrow P(y, z)))$: There are two different students x and y , and if x likes one fast-food, then y likes it as well

Example 6



- Every fast food is liked by some student:

$$\forall y \exists x P(x, y)$$

- All fast food are liked by all students:

$$\forall y \forall x P(x, y)$$

- A fast food is liked by a student:

$$\exists y \exists x P(x, y)$$

- There is a student who likes all fast food:

$$\exists x \forall y P(x, y)$$

Quantifiers Commutativity



- “Similar” quantifiers can be commuted

$$\begin{aligned}\exists x \exists y P(x, y) &= \exists y \exists x P(x, y) \\ \forall x \forall y P(x, y) &= \forall y \forall x P(x, y)\end{aligned}$$

- The mixture of quantifiers should be read from left to right

Quantifiers Commutativity



- “Dissimilar” quantifiers commutation
 1. x, y : integers
 2. $P(x, y): x + y = 0$
- $\forall x \exists y P(x, y)$ - True or False **True**
- $\exists y \forall x P(x, y)$ - True or False **False**
- Dissimilar quantifiers do not commute, in general

Example 7



1. $A(x)$: x lives in Lubbock
2. $B(x, y)$: x and y are friends
3. $D(x)$: x majors in CS
4. Universe of discourse: {All TTU students}

Question: There is a CS major who has a friend living in Lubbock.

$$\exists x(D(x) \wedge \exists y(A(y) \wedge B(x, y)))$$

Example 8(a)



- Express the following statement as a logical expression: “there is a number x such that when it is added to any number, the result is that number, and if it is multiplied by any number, the result is x .” Assume integers as universe of discourse.
- Solution:

$$P(x, y): x + y = y$$

$$Q(x, y): xy = x$$

$$\exists x \forall y (P(x, y) \wedge Q(x, y))$$

Example 8(b)



- Express the following statement as a logical expression: “there is a number x such that when it is added to another number, the result is that (another) number, and if it is multiplied by another number, the result is x .” Assume integers as universe of discourse.
- Solution:

$$P(x, y): x + y = y$$

$$Q(x, y): xy = x$$

$$\exists x \exists y (P(x, y) \wedge Q(x, y))$$

Quantifiers Distributivity



- Universal quantifiers distribute over conjunctions

$$\forall x(P(x) \wedge Q(x)) \equiv \forall xP(x) \wedge \forall xQ(x)$$

- Existential quantifiers distribute over disjunctions

$$\exists x(P(x) \vee Q(x)) \equiv \exists xP(x) \vee \exists xQ(x)$$

- Universal quantifiers do not distribute over disjunctions

$$\forall x(P(x) \vee Q(x)) \not\equiv \forall xP(x) \vee \forall xQ(x)$$

- Existential quantifiers do not distribute over conjunctions

$$\exists x(P(x) \wedge Q(x)) \not\equiv \exists xP(x) \wedge \exists xQ(x)$$

Example 9



$$\neg \forall x (\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$$

Use deMorgan's law and negation of predication

